HRMS AI Chatbot for Odoo HRMS

Project Overview

This project develops an AI-powered HRMS assistant integrated with Odoo to provide instant, accurate answers to employee HR queries. It combines Retrieval-Augmented Generation (RAG), Large Language Models, and company-specific knowledge to reduce HR workload and improve employee experience.

Key Features

• HR policy Q&A; using company documents
• Employee-specific leave, attendance and payroll information
• HR ticket creation for unresolved queries
• Conversation history and secure role-based access
• Source-cited responses to reduce hallucinations

Technology Stack

Frontend: React (planned)
Backend: FastAPI (Python)
LLM: Gemini
RAG: LangChain + LlamaIndex
Embeddings: BAAI bge-small-en-v1.5 (384-dim)
Vector Database: PostgreSQL + pgvector
ORM: SQLAlchemy 2.0
Deployment: Docker

Expected Benefits

• 24/7 HR self-service support
• Faster response times (seconds instead of hours)
• Reduced repetitive HR workload
• Consistent policy guidance with citations
• Scalable architecture for future enterprise growth

Sample project summary prepared from the HRMS AI Chatbot SDLC documentation.